

Problem A. Alice

TimeLimit: 1 second
MemoryLimit: 1024 megabytes

Alice and Bob play a game on a board with n rows and m columns. There is only **one** token on the board, and both players move this same token. It starts on the top-left cell $(1, 1)$, and that cell counts as visited. The players take turns, with Alice moving first.

- On Alice's turn, she moves the token to another cell in the **same row** as its current cell.
- On Bob's turn, he moves the token to another cell in the **same column** as its current cell.

The token may only move to a cell that has never been visited, and the cell it moves to becomes visited. The first player who cannot make a move loses.

Both players play optimally. Determine the winner.

Input

The first line contains an integer T ($1 \leq T \leq 10^5$), the number of test cases. Each of the next T lines contains two integers n and m ($1 \leq n, m \leq 2 \cdot 10^5$).

Output

For each test case, print **Alice** if Alice wins, or **Bob** otherwise.

Example

standard input	standard output
3	Alice
2 2	Alice
3 5	Alice
4 2	

Problem B. Starving Platform

TimeLimit: 1 second
MemoryLimit: 1024 megabytes

The Pit is a vertical prison with n floors, numbered 1 to n from top to bottom, and one prisoner on each floor. The prisoners are never let out. They are fed in a peculiar way.

Every day, a platform loaded with k portions of food starts at floor 1 and slowly descends to floor n , stopping at every floor. The prisoner on a floor may eat from the platform only while it is at their floor, and nobody saves food for the others: everyone eats until their stomach is full or the platform is empty. Whatever is left goes down to the next floor.

The stomach of prisoner i holds at most c_i portions, and it is empty on the first day. Food is never digested: once eaten, it stays in the stomach forever.

A prisoner whose stomach is still empty after the platform has passed goes hungry that day. Prisoner i dies after going hungry on h_i days in a row. Food passes the floor of a dead prisoner untouched.

More precisely, let s_i be the number of portions in the stomach of prisoner i (initially $s_i = 0$). Every day:

1. The platform visits floors $1, 2, \dots, n$ in order. If it arrives at a living prisoner i with r portions, the prisoner eats $\min(c_i - s_i, r)$ of them.
2. Every living prisoner with $s_i = 0$ goes hungry that day. A prisoner who has gone hungry on h_i consecutive days dies.

After enough days, nobody dies anymore. How many prisoners survive forever?

Input

The first line contains two integers n and k ($1 \leq n \leq 2 \cdot 10^5$, $1 \leq k \leq 10^9$): the number of floors and the number of portions on the platform each day.

The i -th of the next n lines describes the prisoner on floor i with two integers c_i and h_i ($1 \leq c_i, h_i \leq 10^9$): the size of their stomach, and how many days in a row they can go hungry before dying.

Output

Print one integer: the number of prisoners who never die.

Examples

standard input	standard output
4 3 5 2 1 2 2 3 4 2	3
5 3 4 1 2 2 3 1 1 4 5 10	4

Note

In the first example, the platform carries 3 portions every day.

- Day 1: prisoner 1 eats all 3 portions. Prisoners 2, 3 and 4 go hungry.
- Day 2: prisoner 1 eats 2 portions, which fills their stomach, and prisoner 2 eats the last portion. Prisoners 3 and 4 go hungry again; prisoner 4 has now gone hungry on $h_4 = 2$ days in a row and dies.
- Day 3: prisoners 1 and 2 are full, so prisoner 3 eats 2 portions and the last portion passes the dead prisoner 4.

From then on, prisoners 1, 2 and 3 always have food in their stomachs, so 3 prisoners survive.

Problem C. Negative Strips

TimeLimit: 1 second
MemoryLimit: 1024 megabytes

Given n , m and k , construct an $n \times m$ integer matrix such that:

- the sum of all elements of the matrix is **positive**;
- the sum of every k consecutive elements in the same row is **negative**;
- the sum of every k consecutive elements in the same column is **negative**.

If $k > m$, a row has no k consecutive elements; if $k > n$, a column has no k consecutive elements.

Input

A single line containing three integers n , m and k ($1 \leq n, m, k \leq 10^6$, $n \cdot m \leq 10^6$).

Output

If no such matrix exists, print No.

Otherwise, print **Yes** on the first line, followed by n lines with m integers each: the matrix. The absolute value of every element must be at most 10^9 . If there are multiple answers, print any of them.

Examples

standard input	standard output
3 5 4	Yes 1 -1 4 -5 1 4 -1 -1 -4 5 1 -4 -1 -1 4
4 5 3	Yes 2 8 -11 -1 11 -5 -3 7 -5 -3 -1 -6 3 2 -9 5 8 -14 2 11
4 6 3	No

Note

In the first example, every 4 consecutive elements in a row have a negative sum, the sum of all elements is 2, and the condition on columns is empty because $k > n$.

In the second example, every 3 consecutive elements in a row or in a column have a negative sum, and the sum of all elements is 1.

Problem D. Piece It Together

TimeLimit: 1 second
MemoryLimit: 1024 megabytes

There was a string S of lowercase English letters. Someone wrote S down n times, and then cut each of the n copies once, splitting it into a non-empty prefix and a non-empty suffix. Different copies may be cut at different positions.

You are given all $2n$ resulting pieces in an **arbitrary** order. Find any string S that could have produced them.

Input

The first line contains an integer n ($n \geq 1$). Each of the next $2n$ lines contains one piece, a non-empty string of lowercase English letters. The total length of all pieces is at most $2 \cdot 10^5$.

It is guaranteed that the pieces were produced from some string S as described.

Output

Print any string S that could have produced the given pieces.

Examples

standard input	standard output
3 ab cab abca b a bcab	abcab
2 ba b b ab	bab

Note

In the first example, the copies of **abcab** were cut as **ab|cab**, **abca|b** and **a|bcab**.

In the second example, the copies of **bab** were cut as **b|ab** and **ba|b**.

Problem E. Steak Sides

TimeLimit: 1 second
MemoryLimit: 1024 megabytes

A steakhouse serves n kinds of steak, numbered 1 to n , and k kinds of side dishes, numbered 1 to k . It offers m set meals. Set meal i contains every steak from l_i to r_i (inclusive), together with one side dish d_i . You want to order some set meals so that every steak $1, 2, \dots, n$ is included in at least one of them. However, you do not want the same side dish twice: **no two ordered set meals may have the same side dish**.

Find the minimum number of set meals to order, or report that it is impossible.

Input

The first line contains three integers n , k and m ($1 \leq n, m \leq 2 \cdot 10^5$, $1 \leq k \leq 20$). The i -th of the next m lines contains three integers l_i , r_i and d_i ($1 \leq l_i \leq r_i \leq n$, $1 \leq d_i \leq k$).

Output

Print the minimum number of set meals, or -1 if it is impossible.

Examples

standard input	standard output
5 3 4 1 4 1 5 5 1 1 2 2 3 5 3	2
3 1 2 1 1 1 2 3 1	-1

Note

In the first example, set meals 1 and 2 would cover all steaks, but both come with side dish 1. Ordering set meals 3 and 4 (or 1 and 4) works.

In the second example, both set meals come with side dish 1, so at most one can be ordered, and neither covers all steaks.

Problem F. Tower Platform

TimeLimit: 1 second
MemoryLimit: 1024 megabytes

In the distant Sky City, there is a path consisting of n high towers. These towers are arranged from left to right, where the position of the i -th tower is (i, h_i) , and h_i represents the height of the i -th tower.

The explorer EZ must travel from the top of the 1st tower to the top of the n -th tower. Since there are no roads between the towers, without any special facilities, he can only move from the top of the i -th tower to the top of the adjacent $(i + 1)$ -th tower. The distance required to move from the top of the i -th tower to the top of the $(i + 1)$ -th tower is defined as the Euclidean distance between the tops of the two towers:

$$\sqrt{(i - (i + 1))^2 + (h_i - h_{i+1})^2}.$$

EZ has obtained **exactly one** special sky platform. If he chooses to build the platform between the i -th and j -th towers, the platform will connect the tops of the two towers in a straight line. EZ can then travel along this platform directly from the top of the i -th tower to the top of the j -th tower without passing through the intermediate towers. The distance required to travel along the platform to the j -th tower is the length of the platform, which is the Euclidean distance between the tops of the two towers:

$$\sqrt{(i - j)^2 + (h_i - h_j)^2}.$$

However, the platform must satisfy the following condition:

The interior of the platform cannot pass through or touch any other towers.

In other words, for all towers k satisfying $i < k < j$, the top of the k -th tower must lie below the platform. Please determine between which two towers the platform should be built to minimize the total distance EZ needs to travel from the top of the 1-st tower to the top of the n -th tower.

Input

The first line contains a single integer n — denotes the number of towers.
The second line contains n integers $h_1, h_2, \dots, h_n$ — denotes height of each tower.

- $2 \leq n \leq 10^6$
- $1 \leq h_i \leq 10^9$

Output

Output a real number representing the minimum total distance required for the explorer EZ to travel from the 1-st tower to the n -th tower under the optimal platform construction.

Your answer will be considered correct if its absolute or relative error does not exceed 10^{-6} . Formally, let your answer be a , the jury's answer be b . Your answer will be considered correct if $\frac{|a-b|}{\max(1, |b|)} \leq 10^{-6}$.

Examples

standard input	standard output
5 1 4 2 5 3	7.63441362
2 1 1000000000	999999999.00000000

Problem G. Edge Flipping

TimeLimit: 1 second
MemoryLimit: 1024 megabytes

You are given an undirected simple graph with n vertices and m edges. Each edge is labeled either 0 or 1. You may perform the following operation any number of times:

- Choose a vertex v .
- Flip the label of every edge incident to v . That is, every label 0 becomes 1, and every label 1 becomes 0.

Determine whether it is possible to make the labels of all edges equal to 0.

Input

The first line contains two integers n and m — the number of vertices and edges. Each of the next m lines contains three integers u , v , and c , indicating that there is an undirected edge between vertices u and v with label c .

The graph is simple: there are no self-loops or multiple edges between the same pair of vertices.

- $2 \leq n \leq 2 \times 10^5$
- $1 \leq m \leq \min\left(2 \times 10^5, \frac{n(n-1)}{2}\right)$
- $1 \leq u, v \leq n, u \neq v$
- $c \in \{0, 1\}$

Output

Print YES if it is possible to make every edge label equal to 0 using the operation any number of times. Otherwise, print NO.

Examples

standard input	standard output
4 4 1 2 1 2 3 0 3 4 1 1 4 0	YES
3 3 1 2 1 2 3 0 1 3 0	NO

Problem H. Host Preliminaries

TimeLimit: 1 second
MemoryLimit: 1024 megabytes

Every year, the National Collegiate Programming Contest (NCPC) holds a preliminary round, and the best teams advance to the final.

Usually, a school's teams take the preliminary round together with other schools. However, if a school's team finishes in the **top three** at the final of year y , the school may host its own preliminary round in each of the next three years: $y + 1$, $y + 2$ and $y + 3$.

In 2024, our school's team “__builtin_orz()” finished third in the final. Therefore, our school may host its own preliminary round in 2025, 2026, and 2027.

You are given every year in which our school's team finished in the top three at the final. In how many different years may our school host its own preliminary round?

Input

The first line contains an integer n ($1 \leq n \leq 2000$), the number of top-three finishes. The second line contains n pairwise distinct integers $y_1, y_2, \dots, y_n$ ($1 \leq y_i \leq 2026$), the years of those finishes. They are not necessarily given in increasing order.

Output

Print one integer: the number of different years in which school may host its own preliminary round.

Example

standard input	standard output
3 11 1 10	7

Note

In the first sample, the finish in year 1 allows hosting in years 2, 3 and 4; the finish in year 10 allows years 11, 12 and 13; and the finish in year 11 allows years 12, 13 and 14. Together, these are the 7 years 2, 3, 4, 11, 12, 13, 14.

Problem I. Three Color Vegetables

TimeLimit: 1 second
MemoryLimit: 1024 megabytes

There are three plates, each containing some green peas, carrot cubes, and corn kernels.
In one operation, you may choose one piece of food from one plate and move it to either of the other two plates.
Your goal is to rearrange the food so that each type of food is placed on a different plate. In other words, after all operations:

- all green peas are on one plate,
- all carrot cubes are on another plate, and
- all corn kernels are on the remaining plate.

The three types of food may be assigned to the three plates in any order. Find the minimum number of operations required.

Input

The input consists of three lines.
The i -th line contains three integers g_i , r_i , and y_i ($0 \leq g_i, r_i, y_i \leq 10^9$), representing the numbers of green peas, carrot cubes, and corn kernels on the i -th plate, respectively.

Output

Print one integer — the minimum number of operations required to rearrange the food so that each type of food is placed on a different plate.

Examples

standard input	standard output
1 2 3 3 1 2 2 3 1	9
1000000000 1000000000 1000000000 1000000000 1000000000 1000000000 1000000000 1000000000 1000000000	6000000000